



PROGRAM
OFFICE

TECHNOLOGY READINESS

TRL 4-6

MATURATION P

■ AUTONOMY ASSURANCE PRO

PROGRAM-OF-RECORD ALIGNMENT BRIEF

CIEMS + AI Mechanic — Tactical UAS Autonomy Assurance Insertion Strategy

PLATFOI

Group 3-4 UAS

DOMA

Autonomous Operations

DAT

June 2026

MATURI

TRL 4-6

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A CRITICAL UNRESOLVED RISK IN FIELDED AUTONOMOUS UAS

Autonomous UAS are flying missions today — without a unified system to verify, monitor, or reconstruct AI decision-making.



RUNTIME MONITORING GAP

No standardized runtime AI behavioral monitoring exists across the Group 3-4 UAS fleet — AI autonomy operates without continuous in-flight oversight.



FORENSIC ACCOUNTABILITY ABSENT

Post-incident forensic capability is absent: AI decisions are effectively unauditably after the fact — leaving commanders, engineers, and oversight authorities with no ground truth.



POLICY COMPLIANCE DEFICIT

DoDD 3000.09 human-machine accountability requirements remain architecturally unsatisfied on fielded platforms — compliance is assumed, not verified.



This is a mission readiness risk and a DoD policy compliance gap — simultaneously.



BEHAVIORAL DRIFT

01

Gradual AI decision-boundary erosion under **environmental distribution shift** — undetectable without continuous monitoring



SENSOR SPOOFING

02

GPS, EO/IR, and radar altimeter manipulation inducing adversarially-directed navigation or targeting errors



MODEL POISONING

03

Supply chain compromise of training data or model weights propagating errors into deployed systems



CONFIDENCE MANIPULATION

04

Adversarial inputs engineered to produce **artificially high model confidence** on incorrect outputs



DECISION CASCADE FAILURES

05

Multi-agent swarm coordination errors amplified across UAS formation — compounding failure propagation

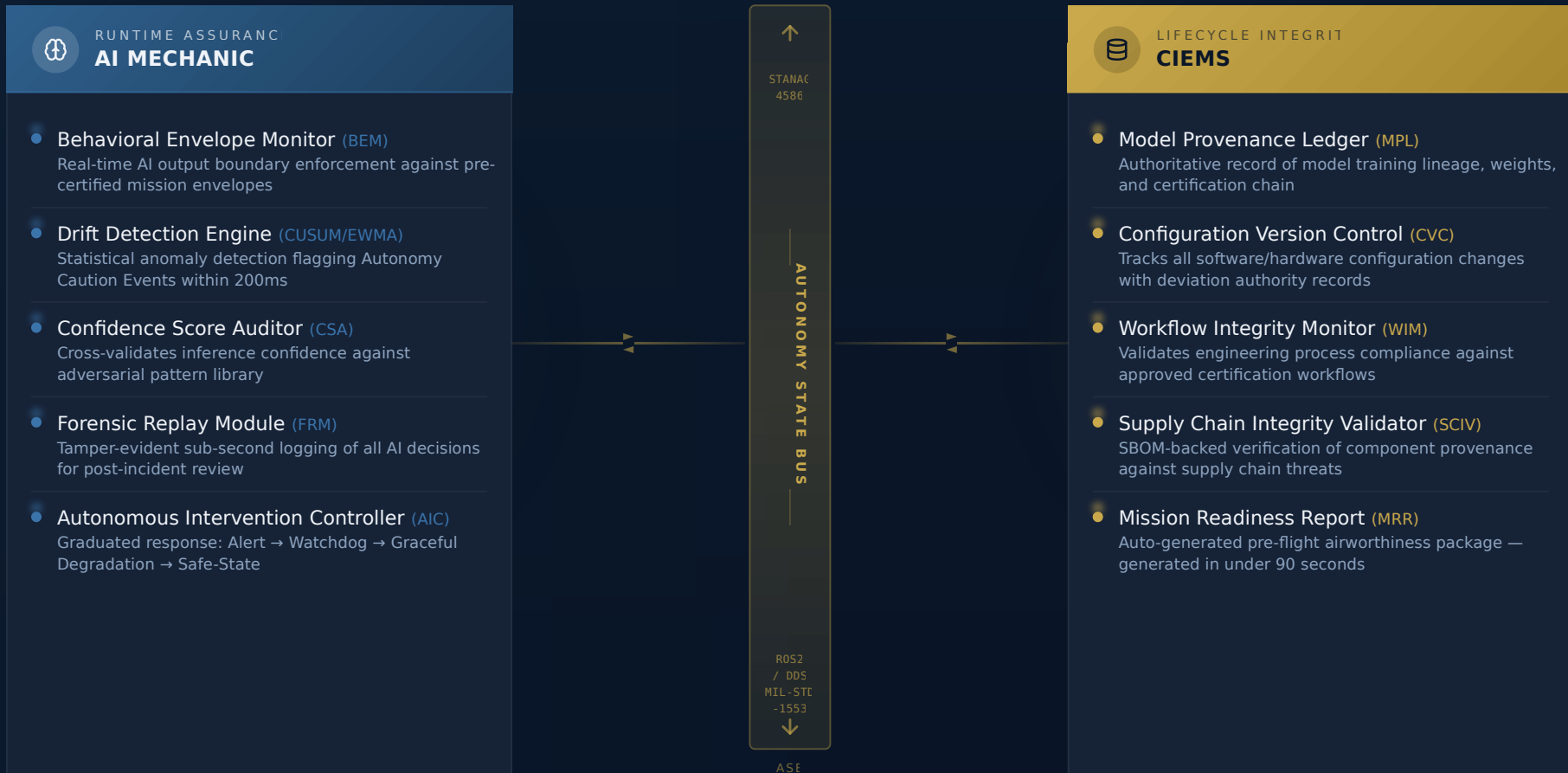


DISTRIBUTION SHIFT

06

Operational environments **outside training data distribution** causing silent AI performance degradation

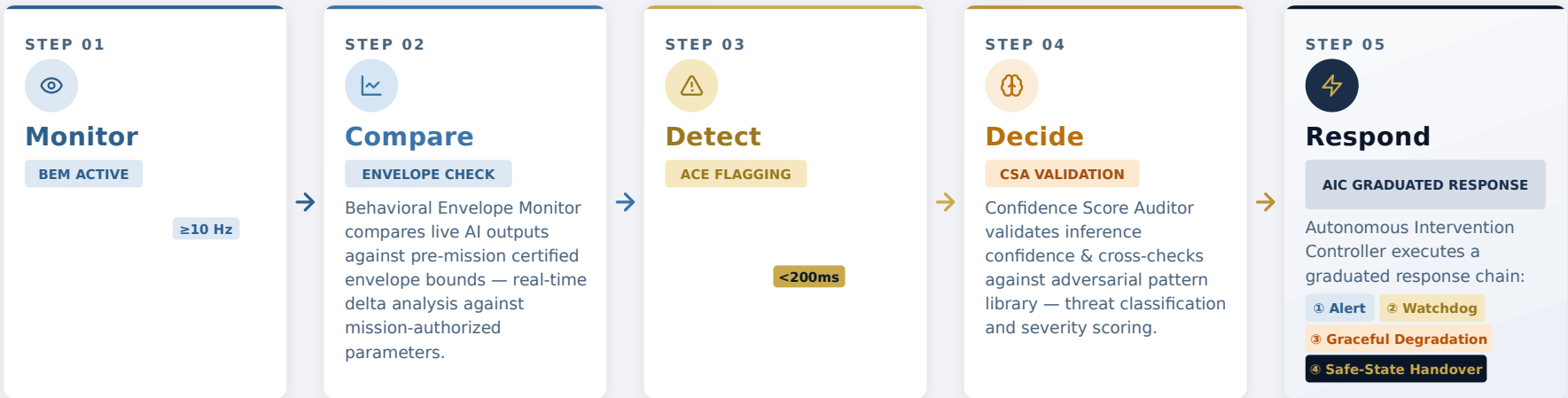
Dual-Pillar Autonomy Assurance: Runtime + Lifecycle



AI Mechanic: Continuous In-Flight Behavioral Oversight

RUNTIME ASSURANCE

FIVE-STAGE CONTINUOUS MONITORING LOOP — ACTIVE ACROSS FULL MISSION DURATION



All events logged to tamper-evident **Forensic Replay Module (FRM)** at sub-second resolution — complete audit trail from sensor input through intervention action, preserved for post-incident forensic review.

LOG RATE

≥10
Hz

Sub-second

Validated KPPs — Autonomy Assurance Performance Envelope

CIEMS + AI Mechanic • Group 3-4 UAS Tactical Autonomy • HWIL-Validated



>97%

ANOMALY DETECTION RA



<200ms

ACE DETECTION LATENCY



<2%

FALSE POSITIVE RA



<500ms

GCS OPERATOR ALERT LATENCY



<2s

SAFE-STATE HANDOVER TIME



<90s

PRE-FLIGHT MRR GENERATION



$\geq 10\text{Hz}$

FORENSIC INFERENCE LOGGING



100%

SBOM COMPONENT COVERAGE



TRL 4→6

MATURATION PACE

! KPPs established against representative mission scenario library; validated through hardware-in-the-loop (HWIL) testing.

HWIL VALIDATED

Platform-Agnostic Integration — No Airframe Modification Required

STANAG 4586 · MIL-STD-1553 · ROS2/DDS

FOUR INTEGRATION PILLARS



CIM

CIEMS Integration Module
Hardware/software payload or GCS rack unit — zero airframe modification.

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-
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ASB

Autonomy State Bus

Connects to **existing avionics** — no new wiring or platform re-cert required.

STANAG 4586

MIL-STD-1553

ROS2/DDS



GCS

GCS Integration

AI Mechanic dashboard **embeds in operator interface** — no new workstation required.

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-
-



CIEMS BACKEND

Classified Cloud + Edge Compute
IL4/IL5 **classified cloud** back-end; edge AI on Jetson Orin-class hardware.

DISA STIG

IL4/IL5

NVIDIA Orin



Airworthiness Safe: Integration does not affect existing platform airworthiness documentation — CIEMS generates an **incremental MRR overlay**, leaving all existing certifications intact and unmodified.

OPERATOR AUTHORITY **PRESERVED** ACROSS ALL AUTONOMY MODES

AI MECHANIC · ADVISORY MODE

All decision authority remains with the operator

- AI Mechanic delivers real-time **advisory alerts** — operator informed, never overridden
- Human retains **all decision authority** throughout mission execution
- CIEMS **documents all mission events** for full accountability record

HUMAN-SUPERVISED AUTONOMOUS

- BEM enforces pre-approved envelope
- ✓ ACE detection: <200ms alert to operator
- ↓ CIEMS logs all ACEs + responses

AIC · AUTONOMOUS SAFE-STATE PROTOCOL

Fail-safe execution — mission integrity preserved

- Upon confirmed critical ACE with no comms: **AIC executes safe-state protocol in <2 seconds**
- All mission data **preserved in tamper-evident FRM** for post-incident forensic review
- Pre-approved intervention rules **eliminate unauthorized autonomous decisions**

LVL 1

HOL

HUMAN-ON-THE-LOOP (HOL)

- Operator remains primary actor
- ✓ DoDD 3000.09 §5 — Fully satisfied
- ⌚ CIEMS audit log: sub-second resolution

LVL 2

HSA

BEM · REAL-TIME ENFORCEMENT

AI executes within pre-approved behavioral envelope

- AI executes within **pre-approved behavioral envelope** — BEM enforces bounds in real time
- Operator **alerted to any deviation** within <200ms via GCS dashboard
- CIEMS logs **all ACEs and system responses** at sub-second forensic resolution

LVL 3

FSA

FAIL-SAFE AUTONOMOUS

- ⚠ Triggered only on confirmed critical ACE
- ⌚ Safe-state handover: <2 seconds KPP
- 📁 FRM preserves 100% forensic record

TIER 1 — MISSION AUTHORITY

Program Office Control

Program office defines **behavioral envelopes**, ROE boundaries, and mission parameters — all encoded pre-flight into the certified autonomy envelope. Mission authority is the authoritative source for permissible AI action space.



TIER 2 — ENGINEERING AUTHORITY

CIEMS Configuration Control

CIEMS manages **configuration control**, deviation authority, and maintains the Model Provenance Ledger as the authoritative engineering record — ensuring integrity from model training through deployment.



TIER 3 — RUNTIME AUTHORITY

AI Mechanic Enforcement Layer

AI Mechanic **enforces behavioral constraints** in real time, logs all anomalies at sub-second resolution, and executes autonomous intervention per pre-approved rules — closing the governance loop at mission execution.

DoDD 3000.09

Human-machine accountability requirements satisfied by design — operator override preserved at all autonomy levels via AIC graduated response architecture.

DoD AI Ethics Principles

Responsible, equitable, traceable, reliable, and governable — all five principles addressed through CIEMS lifecycle integrity and AI Mechanic runtime accountability.

NIST AI RMF (AI 100-1)

GOVERN, MAP, MEASURE, MANAGE functions mapped directly to three-tier governance model — CIEMS provides continuous risk measurement and documentation trail.

MIL-STD-882E

System safety program requirements met via Behavioral Envelope Monitor hazard detection, Drift Detection Engine anomaly flagging, and tamper-evident forensic logging.

MIL-HDBK-516C

Airworthiness certification supported by automated Mission Readiness Report generation — incremental MRR overlay preserves existing platform airworthiness documentation.

JSIG / RMF

CIEMS backend deployed on DISA STIG-compliant IL4/IL5 infrastructure — Supply Chain Integrity Validator and Configuration Version Control directly address JSIG control families.

Phased POR Integration Strategy — 36-Month Execution Plan

TRL 4→6

POR INSERTION

M1 M6 M12 M18 M24 M30 M36

PHASE 1 · MONTHS 1-12

HWIL
ACCEPTANCE
GATE

PHASE 2 · MONTHS 13-24

FLIGHT
DEMO
GATE

PHASE 3 · MONTHS 25-36

PHASE 1 — FOUNDATION

\$4.2M

Architecture & Validation

- Systems architecture finalized & design review completed
- HWIL test rig established with representative UAS avionics
- BEM / DDE / FRM validated against scenario library
- CIEMS backend deployed on classified IL4/IL5 infrastructure

CPFF

PHASE 2 — INTEGRATION

\$9.8M

Platform & Flight Test

- CIM installed on Group 3 testbed aircraft (MQ-1C / RQ-7B)
- GCS dashboard integrated with real-time ACE alert pipeline
- Flight test series initiated — BEM/AIC performance envelope verified
- Airworthiness authority pre-coordination completed

CPFF

PHASE 3 — TRANSITION

\$6.5M

Evaluation & POR Insertion

- Operational evaluation completed — all KPPs demonstrated
- MIL-HDBK-516C compliance package delivered to airworthiness authority
- POR insertion package submitted to program office
- Transition to program-of-record — full fleet deployment authorized

FFP

\$

TOTAL PROGRAM INVESTMENT

\$20.5M

CONTRACT VEHICLE

CPFF → FFP

PHASES 1-2 CONTRACT

Cost-Plus-Fixed-Fee

PHASE 3 CONTRACT

Firm-Fixed-Price



HWIL Gate



Flight Demo Gate

Closing the Autonomy Assurance Gap — Unmatched Capability Integration

CIEMS + AI MECHANIC ✓

CURRENT STATE / ALTERNATIVES ✗

CIEMS + AI MECHANIC

- ✓ **Only solution** integrating runtime monitoring *and* lifecycle integrity in a single coherent architecture
- ✓ Forensic accountability from model training through live flight — **sub-second resolution**
- ✓ **Platform-agnostic**: no airframe modification required
- ✓ Graduated intervention **preserves human authority** without mission abort
- ✓ Automated pre-flight **MRR generation** satisfies airworthiness documentation requirements
- ✓ **STANAG 4586 / ROS2 / MIL-STD-1553 native** — plugs into existing DoD infrastructure

VS

✗ CURRENT STATE / ALTERNATIVES

- ✗ Runtime AI monitoring and lifecycle management are **separate, unintegrated** tools
- ✗ **No standardized forensic AI accountability** capability exists on fielded platforms
- ✗ Point solutions require **platform-specific integration engineering**
- ✗ Binary override systems **force mission abort or full autonomy** — no graduated response
- ✗ Airworthiness documentation is **manual, time-consuming, inconsistent**
- ✗ Custom data bus adapters required — **lengthy integration timelines**



The bottom line: CIEMS + AI Mechanic is the only unified architecture that closes the autonomy assurance gap from training pipeline to live mission — satisfying DoDD 3000.09 accountability requirements without compromising operational tempo.

The Path to **Accountable Autonomous UAS Operations** Starts Here

"**CIEMS + AI Mechanic** delivers the runtime assurance, lifecycle integrity, and governance architecture DoD needs to expand autonomous UAS operations with confidence — across **Group 3-4 platforms**, today."

ACTION



Technical Demonstration Available

HWIL demo of AI Mechanic runtime monitoring on representative UAS avionics — schedule with program team.

ACTION



BAA / OTA Response Ready

Full technical narrative, cost proposal, and compliance matrix packages available for immediate submission.

ACTION



Teaming Discussions Welcome

Open to prime / subcontractor / FFRDC teaming arrangements across all program phases.

POINT OF CONTACT



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CLASSIFICATION
UNCLASSIFIED // FOUO

